FINAL Unit1 MCQ - Topic order - Deduplicated

1. A student wants to create an algorithm to calculate the combined total area of two rectangles. Which of the following algorithms can be used to perform this calculation?

(A)

Step 1:	Multiply the length of the first rectangle by the length of the second rectangle.
Step 2:	Multiply the width of the first rectangle by the width of the second rectangle.
Step 3:	Add the result of step 1 to the result of step 2.

(B)

Step 1:	Multiply the length of the first rectangle by the width of the first rectangle.
Step 2:	Multiply the length of the second rectangle by the width of the second rectangle.
Step 3:	Add the result of step 1 to the result of step 2.

(C)

Step 1:	Add the length of the first rectangle to the width of the first rectangle.
Step 2:	Add the length of the second rectangle to the width of the second rectangle.
Step 3:	Multiply the result of step 1 by the result of step 2.

(D)

Step 1:	Add the length of the first rectangle to the length of the second rectangle.
Step 2:	Add the width of the first rectangle to the width of the second rectangle.
Step 3:	Multiply the result of step 1 by the result of step 2.

2. A carpenter would like to know how many 8-inch-long pieces can be cut from a board whose length is measured in feet. Which of the following algorithms can be used to determine the number of pieces?

(A)

Step 1:	Divide the length of the board by 8.
Step 2:	Keep the whole number portion of the result of step 1, and discard the decimal portion.
Step 3:	Multiply the result of step 2 by 12.

(B)

Step 1:	Divide the length of the board by 12.
Step 2:	Keep the whole number portion of the result of step 1, and discard the decimal portion.
Step 3:	Multiply the result of step 2 by 8.

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(C)

Step 1:	Multiply the length of the board by 8.
Step 2:	Divide the result of step 1 by 12.
Step 3:	Keep the whole number portion of the result of step 2, and discard the decimal portion.

(D)

Step 1:	Multiply the length of the board by 12.
Step 2:	Divide the result of step 1 by 8.
Step 3:	Keep the whole number portion of the result of step 2, and discard the decimal portion.

3. A student wrote a program to calculate the average of several numeric scores. The program compiles without error but produces incorrect results when it runs. The student suspects there is a logic error in the code.

Which of the following strategies is most likely to help identify the logic error?

(A) Changing random parts of the code and rerunning the program until the error is resolved
(B) Inserting print statements throughout the code to check the calculations at various points in the execution
(C) Relying on the compiler's error messages to detect the logic error
(D) Running the program multiple times without making changes to see if the error corrects itself

4. A student wants to create an algorithm to calculate the average (mean) of two quiz scores. Which of the following algorithms can be used to determine the correct average?

(A)

Step one:	Add the two quiz scores.
Step two:	Multiply the result of step one by 2 to determine the average.

(B)

Step one:	Add the two quiz scores.
Step two:	Divide the result of step one by 2 to determine the average.

(C)

Step one:	Subtract the lower quiz score from the higher quiz score.
Step two:	Multiply the result of step one by 2 to determine the average.

(D)

Step one:	Subtract the lower quiz score from the higher quiz score.
Step two:	Divide the result of step one by 2 to determine the average.

5. A student is writing a program that is intended to convert feet to meters. The student enters the code for the program into an integrated development environment (IDE), but the code fails to compile.

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Which of the following steps is most likely to help the student resolve the errors that prevent the code from compiling?

(A) Adding print statements to find and fix any logic errors
(B) Rearranging the order of the statements in the code to fix any sequencing errors
(C) Removing an exception error message by using a different compiler
(D) Using the compiler's error messages to find and fix any syntax errors

6. A teacher wants to calculate the final grade for a student. The final grade is the average of three test scores that have equal weight and one final exam that is weighted twice as much as each test score. The tests and final exam are each scored on a scale of 0 to 100 points.

Which of the following algorithms can be used to determine the correct final grade?

(A)

Step 1:	Compute the sum of the three test scores and the final exam score.
Step 2:	Multiply the result of the step 1 by 2.
Step 3:	Divide the result of step 2 by 5.

(B)

Step 1:	Compute the sum of the three test scores.
Step 2:	Multiply the final exam score by 2 and add it to the result of step 1.
Step 3:	Divide the result of step 2 by 5.

(C)

Step 1:	Compute the sum of the three test scores.
Step 2:	Divide the result of step 1 by 3.
Step 3:	Multiply the final exam score by 2.
Step 4:	Add the results of step 2 and step 3.

(D)

Step 1:	Compute the sum of the three test scores.
Step 2:	Divide the result of step 1 by 2.
Step 3:	Compute the sum of the result of step 2 and the final exam score.
Step 4:	Divide the result of step 3 by 2.

7. A code segment will use variables to represent the age of an animal and whether it is considered a house pet. Which of the following variable declarations is most appropriate for the code segment?

(A)

```
int age;
int pet;
```

(B)

```
int age;
boolean pet;
```

(C)

```
double age;
double pet;
```

(D)

```
boolean age;
boolean pet;
```

8. Consider the following code segment.

```
int x = 5;
int y = 6;
/* missing code */
z = (x + y) / 2;
```

Which of the following can be used to replace /* missing code */ so that the code segment will compile?

(A)

```
boolean z = true;
```

(B)

```
int z = 0.0;
```

(C)

```
int z = 0;
```

(D)

```
z = 0;
```

9. A code segment (not shown) is intended to determine the number of players whose average score in a game exceeds `0.5`. A player's average score is stored in `avgScore`, and the number of players who meet the criterion is stored in the variable `count`.

Which of the following pairs of declarations is most appropriate for the code segment described?

(A)

```
double avgScore;
boolean count;
```

(B)

```
double avgScore;
int count;
```

(C)

```
int avgScore;
boolean count;
```

(D)

```
int avgScore;
int count;
```

10. A code segment will use variables to represent a student's age, in years, and whether the student has a driver's license. Which of the following variable declarations are most appropriate for the code segment?

(A)

boolean age;

double hasLicense;

(B)

boolean age;

boolean hasLicense;

(C)

int age;

boolean hasLicense;

(D)

int age;

double hasLicense;

11. A program needs to maintain the following information about data captured by a sensor.

- Whether the sensor is currently active
- The number of items detected by the sensor
- The average time per detection, in seconds

Which set of variable declarations is most appropriate for this program?

(A)

```
int active;
double items;
boolean avgTime;
```

(B)

```
int active;
int items;
double avgTime;
```

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(C)

```
boolean active;
int items;
boolean avgTime;
```

(D)

```
boolean active;
int items;
double avgTime;
```

12. Consider the following variable declaration.

boolean b;

Which of the following values can be stored in the variable b?

(A)
-1

(B)
0

(C)
1.0

(D)
false

13. Consider the following variable declaration.

int x;

Which of the following values can be stored in the variable x?

(A)
0.5

(B)
1

(C)
true

(D)
false

14. Consider the following code segment.

```
int a = 5;
int b = 8;
int c = 3;
System.out.println(a + b / c * 2);
```

What is printed as a result of executing the code segment?

(A)
2

(B)
6

(C)
8

(D)
9

15. Consider the following code segment.

```
int a = 1;
int b = 2;
int c = 3;
int d = 4;
double x = a + b * c % d;
```

What is the value of x after executing the code segment?

(A)
1.0

(B)
2.5

(C)
3.0

(D)
7.0

16. Consider the following code segment.

```
int x = 10;
int y = 20;
/* missing code */
System.out.print(top / bottom);
```

Which of the following replacements for `/* missing code */` will cause an `ArithmeticException` to occur?

I.
```
int top = x - y;
int bottom = y - x;
```
II.
```
int top = 2 * x;
int bottom = y - top;
```
III.
```
int top = x + y;
int bottom = 2 * top;
```

(A) I only
(B) II only
(C) III only
(D) II and III

17. Consider the following statement.

System.out.print("AP \n Computer \n Science \n A \n rocks");

Which of the following best describes the behavior of this statement?

(A) It prints five lines of text, each separated by a line break indicated by the escape sequence \n.
(B) It prints four lines of text because the escape sequence \n appears four times in the string literal.
(C) It prints one line of text because the printed value is a single string literal.
(D) It prints nothing due to a syntax error.

18. Consider the following code segment.

```
int a = 5;
int b = 2;
double c = 3.0;
System.out.println(5 + a / b * c - 1);
```

What is printed when the code segment is executed?

(A) `9.0`
(B) `10.0`
(C) `11.5`
(D) `14.0`

19. Which of the following arithmetic expressions evaluates to `1` ?

I. `2 / 5 % 3`
II. `2 / (5 % 3)`
III. `2 / 5 + 1`

(A) I only
(B) II only
(C) I and II only
(D) II and III only

20. Consider the following code segment.

```
System.out.print("Hello System.out.println");
System.out.print("!!!");
```

What is printed as a result of executing the code segment?

(A)
```
Hello!!!
```

(B)
```
Hello System.out.println!!!
```

(C)
```
Hello
!!!
```

(D)
```
Hello System.out.println
!!!
```

21. Consider the following code segment.

```
double firstDouble = 2.5;
int firstInt = 30;
int secondInt = 5;
double secondDouble = firstInt - secondInt / firstDouble + 2.5;
```

What value will be assigned to `secondDouble` when the code segment is executed?

(A) `5.0`
(B) `12.5`
(C) `25.5`
(D) `30.5`

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22. Consider the following code segment, which is intended to print the digits of the two-digit `int` number `num` in reverse order. For example, if `num` has the value `75`, the code segment should print `57`. Assume that `num` has been properly declared and initialized.

```
/* missing code */
System.out.print(onesDigit);
System.out.print(tensDigit);
```

Which of the following can be used to replace `/* missing code */` so that the code segment works as intended?

(A)

```
int onesDigit = num % 10;
int tensDigit = num / 10;
```

(B)

```
int onesDigit = num / 10;
int tensDigit = num % 10;
```

(C)

```
int onesDigit = 10 / num;
int tensDigit = 10 % num;
```

(D)

```
int onesDigit = num % 100;
int tensDigit = num / 100;
```

23. Consider the following code segment.

```
System.out.print("AP");
System.out.println();
System.out.println("CS");
System.out.print("A");
```

What is printed as a result of executing the code segment?

(A)
```
APCS
A
```

(B)
```
AP
CSA
```

(C)
```
AP
CS
A
```

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(D)
```
AP

CS
A
```

24. Consider the following code segment.

```
System.out.print(I do not fear computers. );   // Line 1
System.out.println(I fear the lack of them.);  // Line 2
System.out.println(--Isaac Asimov);            // Line 3
```

The code segment is intended to produce the following output but may not work as intended.

```
I do not fear computers. I fear the lack of them.
--Isaac Asimov
```

Which change, if any, can be made so that the code segment produces the intended output?

(A) In line 1, print should be changed to println.
(B) The statement System.out.println() should be inserted between lines 2 and 3.
(C) In lines 1, 2, and 3, the text that appears in parentheses should be enclosed in quotation marks.
(D) No change is needed; the code segment works correctly as is.

25. Consider the following code segment.

```
System.out.print(*);      // Line 1
System.out.print("*");    // Line 2
System.out.println();     // Line 3
System.out.println("*"); // Line 4
```

The code segment is intended to produce the following output, but may not work as intended.

```
**
*
```

Which line of code, if any, causes an error?

(A) Line 1
(B) Line 2
(C) Line 3
(D) The code segment works as intended.

26. Each of the following code segments is intended to print the word `Hello`. Which of the following code segments works as intended?

I. `System.out.print("Hello");`
II. `System.out.print(Hello);`
III. `System.out.print(He);`
`System.out.print(llo);`

(A) I only
(B) III only
(C) I and II
(D) II and III

27. Which of the following code segments will print the word Hello?

(A)
System.out.print("Hello");

(B)
System.out.print(Hello);

(C)
"System.out.print(Hello)";

(D)
System.out.print(He);
System.out.print(llo);

28. Consider the following code segment.

```
System.out.print("*");
System.out.println("**");
System.out.println("***");
System.out.print("****");
```

What is printed as a result of executing the code segment?

(A)
```
*
**
***
****
```

(B)
```
*
**
*******
```

(C)
```
***
***
****
```

(D)
```
***
*******
```

29. Consider the following code segment.

```
System.out.print("One");   // Line 1
System.out.print("Two");   // Line 2
System.out.print("Three"); // Line 3
System.out.print("Four");  // Line 4
```

The code segment is intended to produce the following output, but does not work as intended.

```
OneTwo
ThreeFour
```

Which of the following changes can be made so that the code segment produces the intended output?

(A) Changing `print` to `println` in line 1 only
(B) Changing `print` to `println` in line 2 only
(C) Changing `print` to `println` in line 3 only
(D) Changing `print` to `println` in lines 1, 2, 3, and 4

30. Which of the following statements stores the value `3` in `x` ?

(A) `int x = 7 / 3;`
(B) `int x = 7 / 4;`
(C) `int x = 5 % 8;`
(D) `int x = 8 % 5;`

31. Consider the following code segment.

```
int a = 3 + 2 * 3;
int b = 4 + 3 / 2;
int c = 7 % 4 + 3;
double x = a + b + c;
```

What is the value of `x` after the code segment is executed?

(A)
14.0

(B)
18.0

(C)
20.0

(D)
20.5

32. In the following code segment, `x` and `y` are properly declared and initialized `double` variables.

```
y = x;
double z = y + x;
System.out.println(z);
```

Which of the following best describes what is printed as a result of executing the code segment?

(A) Two times the initial value of `x`
(B) Two times the initial value of `y`
(C) The sum of the initial values of `x` and `y`
(D) The value of `x` followed by the value of `y`

33. Consider the following code segment.

```
int w = 50;
int x = 20;
int y = x;
x = w;
w = x + y;
System.out.println(w + " " + x);
```

What is printed as a result of executing the code segment?

(A)
40 20

(B)
70 20

(C)
70 50

(D)
100 50

34. Consider the following code segment.

```
int first = 5 + 10 * 2;
int second = first + first % 2;
```

What is the value of second after this code segment is executed?

(A)
37

(B)
30

(C)
26

(D)
0

35. Consider the following code segment.

```
int x = 20;
int y = 10;
int temp = x;
x = y;
y = temp * x;
System.out.println(x + " " + y);
```

What is printed as a result of executing this code segment?

(A)
10 100

(B)
10 200

(C)
20 100

(D)
20 200

36. Which of the following expressions evaluates to 3.5?

(A)
(double) 2 / 4 + 3

(B)
(double) 2 / (4 + 3)

(C)
(double) (2 / 4) + 3

(D)
(double) (2 / 4 + 3)

37. Consider the following code segment.

```
double a = 7;
int b = (int) (a / 2);
double c = (double) b / 2;
System.out.print(b);
System.out.print(" ");
System.out.print(c);
```

What is printed as a result of executing the code segment?

(A)
3 1.0

(B)
3 1.5

(C)
3.5 1.5

(D)
3.5 1.75

38. Consider the following code segment.

```
double q = 15.0;
int r = 2;

double x = (int) (q / r);
double y = q / r;

System.out.println(x + " " + y);
```

What is printed as a result of executing this code segment?

(A)
7.0 7.0

(B)
7.0 7.5

(C)
7.5 7.0

(D)
7.5 7.5

39. Consider the following code segment.

```
double a = 1.1;
double b = 1.2;

if ((a + b) * (a - b) != (a * a) - (b * b))
{
  System.out.println("Mathematical error!");
}
```

Which of the following best describes why the phrase `"Mathematical error!"` would be printed?

(Remember that mathematically $(a + b) * (a - b) = a^2 - b^2$.)

(A) Precedence rules make the `if` condition true.
(B) Associativity rules make the `if` condition true.
(C) Roundoff error makes the `if` condition true.
(D) A compiler bug or hardware error has occurred.

40. Consider the following code segment.

```
int a = 5;
int b = 4;
int c = 2;
a *= 3;
b += a;
b /= c;
System.out.print(b);
```

What is printed as a result of executing the code segment?

(A)
4

(B)
9

(C)
9.5

(D)
19

41. The following code segment appears in a method. In the code segment, `y` is an `int` variable.

```
int x = 5 / y;
System.out.println(x);
```

Which of the following preconditions for the method is most appropriate to avoid an ArithmeticException?

(A)
/** Precondition: y is equal to 0. */

(B)
/** Precondition: y is less than 5. */

(C)
/** Precondition: y is not equal to 0. */

(D)
/** Precondition: y is not negative. */

42. The following code segment appears in a method. In the code segment, `x` and `y` are properly declared and initialized `int` variables.

```
System.out.println(x * x * y);
```

Which of the following preconditions for the method is most appropriate to ensure that the value printed by the code segment is always greater than zero?

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(A)

```
/**
 * Preconditions: x is greater than zero.
 *                y is not equal to zero.
 */
```

(B)

```
/**
 * Preconditions: x is not equal to zero.
 *                y is greater than zero.
 */
```

(C)

```
/**
 * Preconditions: x is even and is not equal to zero.
 *                y is not equal to zero.
 */
```

(D)

```
/**
 * Preconditions: x is not equal to zero.
 *                y is even and is not equal to zero.
 */
```

43. In the following code segment, `x` and `y` are properly declared and initialized `int` variables.

```
int z = x;
x = y;
y = z;
```

Which of the following best describes the behavior of the code segment?

(A) It assigns the original value of `x` to both `x` and `y`.
(B) It assigns the original value of `y` to both `x` and `y`.
(C) It leaves the values of `x` and `y` unchanged.
(D) It swaps the values of `x` and `y`.

44. The following code segment contains an error.

```
double pi = 3.14;    // line 1
pi = 3.14159;        // line 2
double other = null; // line 3
double example;      // line 4
example = pi;        // line 5
```

Which of the following best describes the error in this code segment?

(A) In line 2, pi is assigned a value after it was already assigned a value in line 1.
(B) In line 3, null cannot be assigned to a non-reference type.
(C) In line 4, example should be assigned a value.
(D) In line 5, a variable cannot be assigned to another variable.

45. The following code segment is intended to interchange the values of the int variables x and y. Assume that x and y have been properly declared and initialized.

```
int temp = x;
/* missing code */
```

Which of the following can be used to replace /* missing code */ so that the code segment works as intended?

(A)
```
x = y;
y = temp;
```

(B)
```
y = x;
x = temp;
```

(C)
```
y = x;
temp = y;
```

(D)
```
y = x;
temp = x;
```

46. Consider the following code segment.

```
boolean a = true;
boolean b = false;
boolean temp = a;
a = b;
b = temp;
```

```
System.out.println(a);
System.out.println(b);
```

What is printed as a result of executing this code segment?

(A)
true
true

(B)
true
false

(C)
false
true

(D)
false
false

47. The volume of a cylinder is equal to the height times the area of the circular base. The area of the circular base is equal to π (pi) times the square of the radius.
The code segment below is intended to compute and print the volume of a cylinder with radius `r` and height `h`. Assume that the `double` variables `r`, `h`, and `pi` have been properly declared and initialized.

```
/* missing code */
System.out.print(volume);
```

Which of the following can be used to replace `/* missing code */` so that the code segment works as intended?

I.
```
double baseArea = pi * r * r;
double volume = baseArea * h;
```
II.
```
double volume = pi * r * r;
volume = volume * h;
```
III.
```
double volume = pi * r * r * h;
```

(A) I only
(B) I and III only
(C) II and III only
(D) I, II, and III

48. The volume of a cylinder is equal to the height times the area of the circular base. The area of the circular base is equal to π (pi) times the square of the radius.
The following code segment is intended to compute and print the volume of a cylinder with radius r and height h. Assume that the double variables r, h, and pi have been properly declared and initialized.

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```
/* missing code */
System.out.print(volume);
```

Which of the following **cannot** be used to replace /* missing code */ so that the code segment works as intended?

(A)

```
double baseArea;
double volume;
pi * r * r = baseArea;
baseArea * h = volume;
```

(B)

```
double baseArea = pi * r * r;
double volume = baseArea * h;
```

(C)

```
double volume = pi * r * r ;
volume = volume * h;
```

(D)

```
double volume = pi * r * r * h;
```

49. In the following code segment, `x` and `y` are properly declared and initialized `int` variables whose values are positive. The code segment is intended to print the average of `x` and `y`. For example, if `x` is `10` and `y` is `5`, the code segment is intended to print `7.5`.

```
int sum = x + y;
double average = (double) (sum / 2);
System.out.println(average);
```

Which of the following best describes the error, if any, in the code segment?

(A) There is no error; the code segment works as intended.

(B) In the expression `(double) (sum / 2)`, the cast to `double` is applied too late, so the average will be less than the expected result for even values of `sum`.

(C) In the expression `(double) (sum / 2)`, the cast to `double` is applied too late, so the average will be less than the expected result for odd values of `sum`.

(D) In the expression `(double) (sum / 2)`, the cast to `double` is applied too late, so the average will be greater than the expected result for odd values of `sum`.

50. Assume that `x` is a `double` variable with a positive value. Which of the following code segments can be used to round `x` to the nearest integer and store the rounded value in the variable `result`?

(A)
int result = (int) x;

(B)
int result = (int) x + 0.5;

(C)
int result = (int)(x + 0.5);

(D)
int result = (int) x + (int) 0.5;

51. Consider the following code segment.

```
double num = 9 / 4;
System.out.print(num);
System.out.print(" ");
System.out.print((int) num);
```

What is printed as a result of executing the code segment?

(A)
2.0 2

(B)
2.0 2.0

(C)
2.25 2

(D)
2.25 2.0

52. Consider the following code segment.

```
double x = (int) (5.5 - 2.5);
double y = (int) 5.5 - 2.5;
System.out.println(x - y);
```

What is printed as a result of executing the code segment?

(A) `-0.5`
(B) `0.0`
(C) `0.5`

(D) `1.0`

53. Consider the following code segment.

```
int w = 1;
int x = w / 2;
double y = 3;
int z = (int) (x + y);
```

Which of the following best describes the results of compiling the code segment?

(A) The code segment compiles without error.

(B) The code segment does not compile, because the `int` variable `x` cannot be assigned the result of the operation `w / 2`.

(C) The code segment does not compile, because the integer value `3` cannot be assigned to the `double` variable `y`.

(D) The code segment does not compile, because the operands of the addition operator cannot be of different types `int` and `double`.

54. Consider the following code segment.

```
double x = 4.5;
int y = (int) x * 2;
System.out.print(y);
```

What is printed as a result of executing the code segment?

(A) `8`

(B) `8.0`

(C) `9`

(D) `9.0`

55. Consider the following code segment.

```
double w = 2.0;
double x = 5.0;
double y = (int) w / x;
double z = (int) (w / x);

System.out.println(y + ", " + z);
```

What is printed as a result of executing this code segment?

(A)

0.0, 0.0

(B)
0.0, 0.4

(C)
0.4, 0.0

(D)
0.4, 0.4

56. A store sells rope only in whole-foot increments. Given three lengths of rope, in feet, the following code segment is intended to display the minimum length of rope, in feet, that must be purchased so that the rope is long enough to be cut into the three lengths.

For example, for lengths of 2.8 feet, 3 feet, and 5 feet, the minimum length of rope that must be purchased is 11 feet. For these values, the code segment should display 11. As another example, for lengths of 1.1 feet, 3.2 feet, and 2 feet, the minimum length of rope that must be purchased is 7 feet. For these values, the code segment should display 7.

In the following code segment, len1, len2, and len3 are properly declared and initialized double variables representing the three lengths of rope.

```
double total = len1 + len2 + len3;
int minLength = (int) (total + 0.5);
System.out.print(minLength);
```

Which of the following best describes the behavior of the code segment?

(A) The code segment works as intended for all nonnegative values of len1, len2, and len3.
(B) The code segment works as intended but only when the sum of the three lengths is an integer or the decimal part of the sum of the three lengths is greater than or equal to 0.5.
(C) The code segment does not work as intended for some values of len1, len2, and len3 because these values must be cast to int before they are added together.
(D) The code segment does not work as intended for some values of len1, len2, and len3 because minLength should be declared as a double and then cast to an int in the print statement.

57. The following code segment is intended to round `val` to the nearest integer and print the result.

```
double val = -0.7;
int roundedVal = (int) (val + 0.5);
System.out.println(roundedVal);
```

Which of the following best describes the behavior of the code segment?

(A) The code segment works as intended.
(B) The code segment does not work as intended because the expression `(val + 0.5)` should be cast to a `double` instead of an `int`.
(C) The code segment does not work as intended because `val` should be cast to an `int` before `0.5` is added to it.
(D) The code segment does not work as intended because the expression `(int) (val + 0.5)` rounds to the nearest integer only when `val` is positive.

58. The following code segment appears in a method. In the code segment, num1 and num2 are int variables. The method is intended to print the sum of num1 and num2.

```
int result = num1 + num2;
System.out.println(result);
```

Which of the following preconditions for the method is most appropriate to avoid an overflow error?

(A)

```
/** Precondition: num1 and num2 are both positive. */
```

(B)

```
/** Precondition: num1 is not equal to num2 */
```

(C)

```
/** Preconditions: num1 is between Integer.MIN_VALUE
 *            and Integer.MAX_VALUE, inclusive.
 *            num2 is between Integer.MIN_VALUE
 *            and Integer.MAX_VALUE, inclusive.
 */
```

(D)

```
/** Precondition: (num1 + num2) is between Integer.MIN_VALUE
 *           and Integer.MAX_VALUE, inclusive.
 */
```

59. In the code segment below, assume that the `int` variables `a` and `b` have been properly declared and initialized.

```
int c = a;
int d = b;
c += 3;
d--;
double num = c;
num /= d;
```

Which of the following best describes the behavior of the code segment?

(A) The code segment stores the value of `(a + 3) / b` in the variable `num`.
(B) The code segment stores the value of `(a + 3) / (b - 1)` in the variable `num`.
(C) The code segment stores the value of `(a + 3) / (b - 2)` in the variable `num`.
(D) The code segment causes a runtime error in the last line of code because `num` is type `double` and `d` is type `int`.

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60. Consider the following code segment.

```
int x = 0;          // line 1
x++;                // line 2
System.out.print(x);   // line 3
x += 1;             // line 4
System.out.print(x);   // line 5
x *= 2;             // line 6
System.out.print(x);   // line 7
x *= 4;             // line 8
System.out.print(x);   // line 9
```

The code segment is intended to produce the following output. The code segment does not work as intended.

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Which of the following changes can be made so that the code segment works as intended?

(A) Interchanging lines 2 and 4
(B) Interchanging lines 6 and 8
(C) Changing line 6 to `x += 2;`
(D) Changing line 8 to `x *= 2;`

61. Consider the following code segment.

```
int x = 0;
x++;
x += 1;
x = x + 1;
x -= -1;
System.out.println(x);
```

What is printed when the code segment is executed?

(A)
1

(B)
2

(C)
3

(D)
4

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62. Consider the following code segment.

```
num += num;
num *= num;
```

Assume that `num` has been previously declared and initialized to contain an integer value. Which of the following best describes the behavior of the code segment?

(A) The value of `num` is the square its original value.
(B) The value of `num` is two times the square of its original value.
(C) The value of `num` is the square of twice its original value.
(D) It cannot be determined without knowing the initial value of `num`.

63. Consider the following code segment.

```
int x = 5;
x += 6 * 2;
x -= 3 / 2;
```

What is the value of `x` after the code segment is executed?

(A)
1

(B)
9

(C)
15.5

(D)
16

64. Consider the following code segment, where `k` and `count` are properly declared and initialized `int` variables.

```
k++;
k++;
count++;
k--;
count++;
k--;
```

Which of the following best describes the behavior of the code segment?

(A) The code segment leaves both `k` and `count` unchanged.

(B) The code segment increases both `k` and `count` by `2`.

(C) The code segment increases `k` by `4` and `count` by `2`.

(D) The code segment leaves `k` unchanged and increases `count` by `2`.

65. Consider the following code segment.

```
int a = 4;
int b = 5;
a++;
b++;
int c = a + b;
a -= 1;
System.out.println(a + c);
```

What is printed as a result of executing the code segment?

(A)
10

(B)
13

(C)
14

(D)
15

66. Which of the following code segments can be used to print the output 2526?

(A)

```
int val = 5;
val *= val;
System.out.print(val);
val++;
System.out.print(val);
```

(B)

```
int val = 5;
val *= val;
System.out.print(val);
val *= 1;
System.out.print(val);
```

(C)

```
int val = 5;
val += val;
System.out.print(val);
val++;
System.out.print(val);
```

(D)

```
int val = 5;
val += val;
System.out.print(val);
val *= 1;
System.out.print(val);
```

67. The following is an excerpt of a class specification that appears in an API library.

public class Parcel

A Parcel class is used to create objects that represent the status of packages that are delivered to customers.

The Parcel class has status and location variables that hold information about a package's shipping status and location.

The Parcel constructor initializes a Parcel object with its initial status and location.

The updateStatus() and updateLocation() methods are used to update the status and location of a package.

Based on the class specification, which of the following is a true statement about Parcel objects?

(A) status and location are attributes; updateStatus and updateLocation are behaviors.
(B) status and location are behaviors; updateStatus and updateLocation are attributes.
(C) status and updateStatus are attributes; location and updateLocation are behaviors.
(D) status and updateStatus are behaviors; location and updateLocation are attributes.

68. The following are excerpts of two class specifications that appear in an API library.

`PetOwner`

- The `PetOwner` class represents a person who owns a pet that has been a patient at a veterinary office.

- The `PetOwner` class has a `String` variable, `name`, that holds information about the person's name and an `int` variable, `numOfPets`, that represents the number of pets the person owns.
- The `getPersonName()` and `getNumOfPets()` methods are used to get information about the pet owner.

`Cat`

- The `Cat` class represents information about a cat that is a patient at a veterinary office.
- The `Cat` class has a `String` variable, `catName`, that holds information about the cat's name and a `PetOwner` variable, `owner`, that holds information about the cat's owner.
- The `getCatOwner()` and `getCatName()` methods are used to get information about the cat.

Based on the class specifications, which of the following descriptions is most accurate?

(A) A `Cat` object is an attribute of a `PetOwner` object.
(B) A `PetOwner` object is an attribute of a `Cat` object.
(C) The method `getCatName` is a behavior of a `PetOwner` object.
(D) The method `getNumOfPets` is an attribute of a `PetOwner` object.

69. The following is an excerpt of a class specification that appears in an API library.

public class Robot

A Robot specifies a robot that moves around a two-dimensional coordinate grid.

The Robot class has xCoordinate and yCoordinate variables that hold information about the robot's location on the grid.

The Robot constructor initializes a Robot object with the given coordinates.

The move() and rotate() methods are used to modify the robot's location on the grid.

Based on the class specification, which of the following descriptions is accurate?

(A) Robot objects are attributes of the methods move and rotate.
(B) Robot objects are behaviors of the methods move and rotate.
(C) The methods move and rotate represent attributes of Robot objects.
(D) The methods move and rotate represent behaviors of Robot objects.

70. The following is an excerpt of a class specification that appears in an API library.

`public class Shirt`

The `Shirt` class is used to create objects that represent shirts.

The `Shirt` class has `color` and `size` variables that hold information about the characteristics of a shirt.

The `Shirt` constructor initializes a `Shirt` object with the given color and size.

The `getColor` and `getSize` methods return information about a shirt.

A `poloShirt` object will be declared as type `Shirt`.

Based on the class specification, which of the following descriptions is accurate?

(A) The `color` and `size` variables are attributes of the `poloShirt` object.
(B) The `color` and `size` variables are behaviors of the `poloShirt` object.

(C) The `poloShirt` object is an attribute of the `getColor` and `getSize` methods.
(D) The `poloShirt` object is a behavior of the `getColor` and `getSize` methods.

71. The following is an excerpt of a class specification that appears in an API library.

`public class Song`
A `Song` specifies a piece of music stored in a playlist.
The `Song` class has `title` and `artist` variables that hold information about a song.
The `Song` constructor initializes a `Song` object with the given title and artist.
The `play()` and `stop()` methods are used to control playback of the song.

Based on the class specification, which of the following correctly identifies one attribute and one behavior of the `Song` class?

(A) Attribute: `title`
Behavior: `play()`
(B) Attribute: `play()`
Behavior: `artist`
(C) Attribute: `Song` constructor
Behavior: `artist`
(D) Attribute: `Song` constructor
Behavior: `stop()`

72. The following is an excerpt of a class specification that appears in an API library.

public class Vehicle
The Vehicle class is used to create objects that represent vehicles.
The Vehicle class has make, model, and year variables that hold information about the vehicle's characteristics.
The Vehicle constructor initializes a Vehicle object with the given make, model, and year.
The getMake(), getModel(), and getYear() methods return information about the vehicle.

Based on the class specification, which of the following descriptions is accurate?

(A) Vehicle objects are attributes of the variables make, model, and year.
(B) Vehicle objects are behaviors of the variables make, model, and year.
(C) The variables make, model, and year are attributes of Vehicle objects.
(D) The variables make, model, and year are behaviors of Vehicle objects.

73. A school administrator has created a Student class. The class contains variables to represent the following.

- An int variable named studentID to represent the student's ID number
- A String variable named studentName to represent the student's name

The school administrator has also created a Parent class. The class contains variables to represent the following.

- A String variable named parentName to represent the parent's name
- A String variable named email to represent the parent's e-mail address

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The object penelope will be declared as type Student. The object mrsPatel will be declared as type Parent.

Which of the following descriptions is accurate?

(A) An attribute of the penelope object is email.
(B) An attribute of the penelope object is Parent.
(C) An attribute of the penelope object is Student.
(D) An attribute of the mrsPatel object is email.

74. A teacher has created a Student class. The class contains the following.

- An int variable named grade to represent the student's grade level
- A String variable named name to represent the student's name
- A double variable named average to represent the student's grade point average
- A method named updateAverage that updates the value of average.

The object greg will be declared as type Student.

Which of the following descriptions is accurate?

(A) greg is an instance of the Student class.
(B) greg is an instance of three attributes.
(C) Student is an instance of the greg object.
(D) updateAverage is an instance of the Student class.

75. A method has a precondition that the variable `num` must be a positive integer divisible by `2` or must be greater than `100`. Which of the following values of `num` does **not** satisfy the precondition?

(A)
```
4
```

(B)
```
5
```

(C)
```
104
```

(D)
```
105
```

76. The following code segment appears in a method. The method is intended to print the product of 3 and the nonnegative difference between the int variables num1 and num2.

```
System.out.println(3 * (num1 - num2));
```

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Which of the following preconditions, if any, is required so that the method works as intended?

(A)
```
/** Preconditions: num1 >= 0, num2 >= 0 */
```

(B)
```
/** Precondition: num1 >= num2 */
```

(C)
```
/** Precondition: num2 >= num1 */
```

(D) No precondition is required.

77. The following code segment appears in a method. In the code segment, x and y are double variables.

```
double result = (x - y) / 2.0;
System.out.println(result);
```

Which of the following preconditions for the method is most appropriate to ensure that the value printed by the code segment is always positive?

(A)
```
/** Precondition: x and y are both positive. */
```

(B)
```
/** Precondition: x and y are both even. */
```

(C)
```
/** Precondition: x > y */
```

(D)
```
/** Precondition: x < y */
```

78. Consider the following code segment.

```
double result = Math.abs(x);
```

Which of the following statements about the variable x is true?

(A) x may hold any numerical value.
(B) x must be equal to 0.
(C) x must have a value greater than 0.
(D) x must have a value less than 0.

79. Consider the following method.

```
public double myMethod(int a, boolean b)
{  /* implementation not shown */  }
```

Which of the following lines of code, if located in a method in the same class as `myMethod`, will compile without error?

(A) `int result = myMethod(2, false);`
(B) `double result = myMethod(0, false);`
(C) `double result = myMethod(true, 10);`
(D) `double result = myMethod(2.5, true);`

80. The `Student` class has been defined to store and manipulate grades for an individual student. The following methods have been defined for the class.

```
/* Returns the sum of all of the student's grades */
public double sumOfGrades()
{  /* implementation not shown */  }
/* Returns the total number of grades the student has received */
public int numberOfGrades()
{  /* implementation not shown */  }
/* Returns the lowest grade the student has received */
public double lowestGrade()
{  /* implementation not shown */  }
```

Which of the following statements, if located in a method in the `Student` class, will determine the average of all of the student's grades except for the lowest grade and store the result in the `double` variable `newAverage`?

(A) `newAverage = sumOfGrades() / (numberOfGrades() - 1);`
(B) `newAverage = sumOfGrades() - lowestGrade() / (numberOfGrades() - 1);`
(C) `newAverage = (sumOfGrades() - lowestGrade()) / numberOfGrades() - 1;`
(D) `newAverage = (sumOfGrades() - lowestGrade()) / (numberOfGrades() - 1);`

81. Consider the following method.

```
public void doSomething()
{
System.out.println("Something has been done");
}
```

Each of the following statements appears in a method in the same class as `doSomething`. Which of the following statements are valid uses of the method `doSomething`?

I. `doSomething();`
II. `String output = doSomething();`
III. `System.out.println(doSomething());`

(A) I only

(B) I and II only
(C) I and III only
(D) I, II, and III

82. Consider the following class definition.

```
public class ExamScore
{
private String studentId;
private double score;
public ExamScore(String sid, double s)
{
    studentId = sid;
    score = s;
}
public double getScore()
{
    return score;
}
public void bonus(int b)
{
    score += score * b/100.0;
}
}
```

Assume that the following code segment appears in a class other than `ExamScore`.

```
ExamScore es = new ExamScore("12345", 80.0);
es.bonus(5);
System.out.println(es.getScore());
```

What is printed as a result of executing the code segment?

(A) `4.0`

(B) `80.0`

(C) `84.0`

(D) `85.0`

83. Consider the following description of method `fahrenheitToCelsius`.
`public double fahrenheitToCelsius(double f)` – Takes as input a temperature in degrees Fahrenheit and returns the corresponding temperature in degrees Celsius.

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Assume that each of the following code segments appears in a method in the same class as `fahrenheitToCelsius`. Which of the following code segments prints the temperature in degrees Celsius that corresponds to 32 degrees Fahrenheit?

(A)
```
double f = 32.0;
f = fahrenheitToCelsius();
System.out.println(f);
```

(B)
```
double f = 32.0;
fahrenheitToCelsius(f);
System.out.println(f);
```

(C)
```
double f = 32.0;
double c = fahrenheitToCelsius();
System.out.println(c);
```

(D)
```
double f = 32.0;
double c = fahrenheitToCelsius(f);
System.out.println(c);
```

84. Consider the following descriptions of two methods that appear in the same class.

Method Signature	Explanation
public void methodA(int arg)	Calls methodB with the value of arg * 10
public void methodB(int arg)	Displays the value of arg + 10

Consider the call methodA(4), which appears in a method in the same class. What, if anything, is printed as a result of the call methodA(4)?

(A)
40

(B)
50

(C)
140

(D) Nothing is printed.

85. Consider the following description of method `min`.
`public int min(int first, int second)` – Returns the lesser of its two parameters.

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Assume that each of the following expressions appears in a method in the same class as the method `min`. Assume also that the `int` variables `p`, `q`, and `r` have been properly declared and initialized. Which of the following expressions evaluates to the minimum value among `p`, `q`, and `r`?

I. `min(min(p, q), r)`
II. `min(p, min(q, r))`
III. `min(min(p, q), p)`

(A) I only
(B) II only
(C) I and II only
(D) I, II, and III

86. Consider the following description of method printSomething.

Method Signature	Explanation
public void printSomething(int num, boolean val)	Prints the value of val immediately followed by the value of num - 1.

Consider the following code segment, which appears in the same class as printSomething.

```
printSomething(1, true);
printSomething(2, true);
```

What is printed as a result of executing the code segment?

(A)
0true1true

(B)
1true2true

(C)
true0true1

(D)
true1true2

87. Consider the following method signatures.

Method Signature	Explanation
public void printSum(int x, double y)	Prints the sum of `x` and `y`

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Method Signature	Explanation
public void printProduct(double x, int y)	Prints the product of `x` and `y`

Consider the following code segment.

```
int num1 = 5;
double num2 = 10.0;
printSum(num1, num2);
System.out.println();
printProduct(num1, num2);
```

What, if anything, is printed as a result of executing the code segment?

(A)

15

50

(B)

15

50.0

(C)

15.0

50.0

(D) Nothing is printed because the code does not compile.

88. Consider the following description of method printValues.

Method Signature	Explanation
public void printValues(int numTimes, int val)	Prints the value of val a total of numTimes times, then moves the cursor to a new line.

Consider the following code segment, which appears in the same class as printValues.

```
printValues(2, 3);
printValues(4, 5);
```

What is printed as a result of executing the code segment?

(A)

222

44444

(B)

2222

33333

(C)

33

5555

(D)

44

555

89. Consider the following description of method showResult.

Method Signature	Explanation
public void showResult(int x, int y)	Prints the value of y - x

Consider the following code segment, which appears in the same class as showResult.

```
showResult(1, 3);
System.out.print(" ");
showResult(4, 10);
```

What is printed as a result of executing the code segment?

(A)

-6 -2

(B)

-2 -6

(C)

2 6

(D)

6 2

90. Consider the following method signatures and corresponding descriptions. Assume that both methods appear in the same class.

Method Signature	Description
public int methodOne(int i, int j)	Returns the value of `i + j`
public int methodTwo(int m, int n)	Returns the value of `n - m`

Consider the following code segment, which appears in the same class as `methodOne` and `methodTwo`.

```
int x = methodOne(4, 5) * methodTwo(1, 3);
System.out.println(x);
```

What is printed as a result of executing the code segment?

(A)

-18

(B)

-4

(C)

4

(D)

18

91. Consider the following description of the `MagicNumber` class which includes one constructor and two methods.

`public MagicNumber()` – Constructs a `MagicNumber` object that represents a number

`public void displayNumber()` – Displays the number to the screen

`public void add_2()` – Increases the number by 2

When located in a method in a class other than `MagicNumber`, which of the following code segments will compile without error?

I.
```
MagicNumber.add_2();
MagicNumber.displayNumber();
```

II.
```
MagicNumber n1 = new MagicNumber();
n1.add_2();
n1.displayNumber();
```

III.
```
n2.add_2();
n2.displayNumber();
```

(A) I only
(B) II only
(C) II and III only
(D) None of the code segments will compile.

92. Consider the following description of the `Purchase` class which includes one constructor and one method.

`public Purchase(double purchaseAmt, double taxAmt)` – Constructs a `Purchase` object that represents a single purchase with purchase amount `purchaseAmt` and tax `taxAmt`.

`public void totalAmount()` – Displays the sum of the purchase amount and the tax

Assume that a `Purchase` object `p` has been properly declared and initialized. Which of the following code segments will successfully print the total purchase amount associated with `p`?

(A) `Purchase.totalAmount();`
(B) `System.out.print(p);`
(C) `System.out.print(p.totalAmount());`
(D) `p.totalAmount();`

93. Consider the following Painting class.

```
public class Painting
{
  private int year;
  private String artist;
  private String title;

  public Painting(int y, String a)
  {
    year = y;
    artist = a;
    title = "Untitled";
  }

  // There are no other constructors.
}
```

Which of the following code segments, appearing in a class other than Painting, will correctly create an instance of a Painting object?

(A)
```
Painting p = new Painting("Frida Kahlo", "Self Portrait", 1939);
```

(B)
```
Painting p = new Painting(1939, "Frida Kahlo", "Self Portrait");
```

(C)

```
Painting p = new Painting("Frida Kahlo", 1939);
```

(D)

```
Painting p = new Painting(1939, "Frida Kahlo");
```

94. Consider the following Date class.

```
public class Date
{
  private String month;
  private int day;

  public Date(String m, int d)
  { /* implementation not shown */ }

  // Class contains no other constructors. Other methods are not shown.
}
```

Which of the following code segments, appearing in a class other than Date, will correctly create an instance of a Date object?

(A)

```
Date birthday = new Date("September", "5th");
```

(B)

```
Date birthday = new Date("September", 5);
```

(C)

```
Date birthday = new Date("September 5");
```

(D)

```
Date birthday = new Date();
birthday.month = "September";
birthday.day = 5;
```

95. The following method header appears in the `Quadrilateral` class.

```
/**
 *  Returns the area of a quadrilateral with
 *  length len and width wid
 */
public static double findArea(double len, double wid)
```

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Which of the following code segments, when appearing in a class other than `Quadrilateral`, will compile without error?

(A)

```
int area = findArea(11.0, 8.5);
```

(B)

```
double area = findArea(11.0, 8.5);
```

(C)

```
int area = Quadrilateral.findArea(11.0, 8.5);
```

(D)

```
double area = Quadrilateral.findArea(11.0, 8.5);
```

96. Consider the following information about the calculateBill method.

- The method is a static method in the Store class.
- The method has one int parameter.
- The method has return type double.
- The method can be accessed from another class.

Which of the following code segments, when appearing in a method in a class other than Store, will compile without error?

(A)

```
double tot = calculateBill(5);
```

(B)

```
double tot = Store.calculateBill(5);
```

(C)

```
double tot;
tot.calculateBill(5);
```

(D)

```
double tot = 5.0;
tot.calculateBill();
```

97. Consider the following information about the getTax method.

- The method is a static method in the PurchaseManager class.
- The method has one double parameter.

- The method returns a double equal to its parameter times 0.08.

The following code segment appears in another method in the PurchaseManager class.

```
double tax = getTax(100.0);
System.out.println(tax);
```

What, if anything, is printed as a result of executing the code segment?

(A)
8.0

(B)
108.0

(C) Nothing is printed because the method call should be PurchaseManager.getTax(100.0).

(D) Nothing is printed because the return value from getTax cannot be assigned to the variable tax.

98. Consider the following information about the generatePasscode method.

- The method appears in the AccountManager class.
- The method has no parameters.
- The method has return type int.
- The method can be accessed from another class.

The following code segment appears in a method in a class other than AccountManager class.

```
int c = AccountManager.generatePasscode();
System.out.println(c);
```

Which of the following must be true for the code segment to compile without error?

(A) The generatePasscode method should be designated as void.

(B) The generatePasscode method should be designated as static.

(C) The AccountManager class should contain another method with the name generatePasscode that has at least one parameter.

(D) The AccountManager class should contain another method with the name generatePasscode that has return type double.

99. The `toCelsius` method appears in the `TemperatureConverter` class. The method takes a parameter that represents a temperature in degrees Fahrenheit and correctly returns the equivalent temperature in degrees Celsius. The header for the `toCelsius` method is shown.

```
/**
 *  Returns the Celsius equivalent of a
```

```
 *  Fahrenheit temperature
 */
public static double toCelsius(double fahrenheit)
```

The following code segment appears in a class other than `TemperatureConverter`. It is intended to print the Celsius equivalent of 212 degrees Fahrenheit.

```
double temp = toCelsius(212.0);  // Line 1
System.out.println(temp);
```

Which of the following changes, if any, can be made so that the code segment works as intended?

(A) No change is needed; the code segment works as intended.
(B) Changing line 1 to `double temp.toCelsius(212.0);`
(C) Changing line 1 to `double temp = TemperatureConverter.toCelsius(212.0);`
(D) Changing line 1 to `double temp = toCelsius.TemperatureConverter(212.0);`

100. Consider the following description of the `AnimalPrinter` class which includes two methods.

`public void printDog()` – Displays the word "`dog`" and then moves the cursor to a new line

`public void printCat()` – Displays the word "`cat`" and then moves the cursor to a new line

The method `myMethod` appears in a class other than `AnimalPrinter`. The method is intended to produce the following output.

```
dog
cat
```

Assume that an `AnimalPrinter` object `myPrinter` has been properly declared and initialized inside `myMethod`. Which of the following code segments, if located in `myMethod`, will produce the intended output?

(A)
```
printDog();
printCat();
```
(B)
```
printDog(AnimalPrinter);
 printCat(AnimalPrinter);
```
(C)
```
AnimalPrinter.printDog();
 AnimalPrinter.printCat();
```
(D)
```
myPrinter.printDog();
myPrinter.printCat();
```

101. The AnimalPrinter class includes the following two methods.

Method Signature	Explanation
public void printDog()	Prints the word "dog" and then moves the cursor to a new line.

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Method Signature	Explanation
public void printCat()	Prints the word "cat" and then moves the cursor to a new line.

The method myMethod appears in a class other than AnimalPrinter. The method is intended to produce the following output.

dog
cat

Assume that an AnimalPrinter object myPrinter has been properly declared and initialized inside myMethod. Which of the following code segments, if located in myMethod, will produce the intended output?

(A)

```
printDog();
printCat();
```

(B)

```
printDog(AnimalPrinter);
printCat(AnimalPrinter);
```

(C)

```
AnimalPrinter.printDog();
AnimalPrinter.printCat();
```

(D)

```
myPrinter.printDog();
myPrinter.printCat();
```

102. Consider the following class definition.

```
public class Thing
{
 public void talk()
 {
     System.out.print("Hello ");
 }

 public void name()
 {
     System.out.print("my friend");
 }
```

```
    public void greet()
    {
        talk();
        name();
    }
    /* Constructors not shown */
}
```

Which of the following code segments, if located in a method in a class other than `Thing`, will cause the message `"Hello my friend"` to be printed?

(A)
```
Thing a = new Thing();
Thing.talk();
Thing.name();
```

(B)
```
Thing a = new Thing();
Thing.greet();
```

(C)
```
Thing a = new Thing();
a.greet();
```

(D)
```
Thing a = new Thing();
a.name();
a.talk();
```

103. Consider the following method, which is intended to return true if the distance between d1 and d2 is no more than tolerance units, and to return false otherwise.

For example, almostEqual(7.9, 7.8, 0.15) should return true because the distance between 7.9 and 7.8 is less than 0.15, and almostEqual(54.124, 54.126, 0.001) should return false because the distance between 54.124 and 54.126 is greater than 0.001.

```
public static boolean almostEqual(double d1, double d2,
                                  double tolerance)
{
  /* missing code */;
}
```

Which of the following can be used to replace /* missing code */ so that the method will work as intended?

(A)
```
return (d1 - d2) <= tolerance
```

(B)
```
return ((d1 + d2) / 2) <= tolerance
```

(C)
```
return Math.abs(d1 - d2) <= tolerance
```

(D)

return Math.abs((d1 + d2) / 2) <= tolerance

104. Consider the following statement.

int b = 5 + (int) (Math.random() * 5);

What are the possible values that can be assigned to `b` as a result of executing the statement?

(A) 5, 6, 7, 8, or 9
(B) 5, 6, 7, 8, 9, or 10
(C) 6, 7, 8, or 9
(D) 6, 7, 8, 9, or 10

105. Consider the following statement, which is intended to assign the value of the expression $\sqrt{\frac{(x+y)^2}{|a-b|}}$ to the variable `result`. Assume that the `double` variables `x`, `y`, `a`, and `b` have been properly declared and initialized.

double result = /* missing code */ ;

Which of the following can replace `/* missing code */` so that the statement works as intended?

(A)

Math.sqrt((x + y) ^ 2 / Math.abs(a, b))

(B)

Math.sqrt((x + y) ^ 2 / Math.abs(a - b))

(C)

Math.sqrt(Math.pow(x + y, 2) / Math.abs(a, b))

(D)

Math.sqrt(Math.pow(x + y, 2) / Math.abs(a - b))

106. Which of the following statements assigns a random integer between 25 and 60, inclusive, to `rn` ?

(A) `int rn = (int) (Math.random() * 25) + 36;`
(B) `int rn = (int) (Math.random() * 25) + 60;`
(C) `int rn = (int) (Math.random() * 36) + 25;`
(D) `int rn = (int) (Math.random() * 60) + 25;`

107. Which of the following statements assigns a random integer between `1` and `25` inclusive, to `rand`?

(A)

```
int rand = (int) (Math.random()) * 25 + 1;
```

(B)

```
int rand = (int) (Math.random() * 25);
```

(C)

```
int rand = (int) (Math.random() * 25) + 1;
```

(D)

```
int rand = (int) (Math.random() + 1) * 25;
```

108. Consider the following code segment, which is intended to assign to `num` a random integer value between `min` and `max`, inclusive, each with an equal chance of being assigned to `num`. Assume that `min` and `max` are integer variables and that the value of `max` is greater than the value of `min`.

```
double rn = Math.random();
int num = /* missing code */ ;
```

Which of the following can replace `/* missing code */` so that the code segment works as intended?

(A)

```
(int) (rn * max) + min
```

(B)

```
(int) (rn * (max - min)) + min
```

(C)

```
(int) (rn * (max - min)) + 1
```

(D)

```
(int) (rn * (max - min + 1)) + min
```

109. Consider the following code segment.

```
int x = ((int) (Math.random() * 10)) + 5;
int y = ((int) (Math.random() * 5)) + 10;
System.out.println(x + " " + y);
```

Which of the following could be printed as a result of executing this code segment?

(A)

```
8 15
```

(B)
9 12

(C)
10 7

(D)
15 11

110. Consider the following code segment.

```
int num = (int)(Math.random() * 6);
System.out.println(num);
```

Which of the following **cannot** be printed as a result of executing this code segment?

(A)
0

(B)
2

(C)
4

(D)
6

111. Consider the following code segment. Assume that `a` is greater than zero.

```
int a = /* value not shown */;
int b = a + (int) (Math.random() * a);
```

Which of the following best describes the value assigned to `b` when the code segment is executed?

(A) `a`
(B) A random integer between `0` and `a - 1`, inclusive
(C) A random integer between `a` and `2 * a`, inclusive
(D) A random integer between `a` and `2 * a - 1`, inclusive

112. Consider the following statement, which is intended to generate an integer between `1` and `10`, inclusive, and assign it to `result`. This statement does not work as intended.

```
int result = (int) Math.random() * 10 + 1;
```

Which of the following changes can be made so that this statement works as intended?

(A) Add parentheses around the expression 10 + 1, as shown in the following statement.

```
int result = (int) Math.random() * (10 + 1);
```

(B) Add parentheses around the expression Math.random() * 10, as shown in the following statement.

```
int result = (int) (Math.random() * 10) + 1;
```

(C) Change the value of 10 to 9, as shown in the following statement.

```
int result = (int) Math.random() * 9 + 1;
```

(D) Remove the cast of Math.random() to an int value, as shown in the following statement.

```
int result = Math.random() * 10 + 1;
```

113. Assume that the following variable declarations have been made.

```
double d = Math.random();
double r;
```

Which of the following assigns a value to `r` from the uniform distribution over the range `0.5 ≤ r < 5.5` ?

(A) `r = d + 0.5;`
(B) `r = d + 0.5 * 5.0;`
(C) `r = d * 5.0;`
(D) `r = d * 5.0 + 0.5;`

114. Consider the following statement.

```
int r = (int) (Math.random() * 6) + 10;
```

Which of the following best describes the behavior of the statement?

(A) It generates a random integer between 6 and 15, inclusive, and assigns it to r.
(B) It generates a random integer between 6 and 16, inclusive, and assigns it to r.
(C) It generates a random integer between 10 and 15, inclusive, and assigns it to r.
(D) It generates a random integer between 10 and 16, inclusive, and assigns it to r.

115. The Dog class has a constructor that takes two parameters. The first parameter is a String representing a dog's name and the second parameter is an int representing the dog's age, in months.

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The following code segment appears in a class other than Dog.

```
Dog dog1 = new Dog("Rex", 4);
Dog dog2 = dog1;
dog1 = new Dog("Fido", 60);
```

Which of the following best describes the contents of dog1 and dog2 as a result of executing this code segment?

(A) dog1 and dog2 are Dog variables that each contain the same Dog object, which represents a dog named "Fido" that is 60 months old.

(B) dog1 is a Dog variable that contains a Dog object representing a dog named "Fido" that is 60 months old, and dog2 is a Dog variable that contains a Dog object representing a dog named "Rex" that is 4 months old.

(C) dog1 and dog2 are Dog reference variables that each contain a reference to the same memory address that contains the Dog object representing a dog named "Fido" that is 60 months old.

(D) dog1 is a Dog reference variable, which contains a reference to the memory address for the Dog object representing a dog named "Fido" that is 60 months old. dog2 is a Dog reference variable, which contains a reference to the memory address for the Dog object representing a dog named "Rex" that is 4 months old.

116. A student has created a `Car` class. The class contains the following attributes.

- A `String` variable named `color` to represent the color of the car
- An `int` variable named `year` to represent the year the car was made
- A `String` variable named `make` to represent the manufacturer of the car
- A `String` variable named `model` to represent the model of the car

Assume that `vehicle` is a variable of type `Car` that has been properly declared and initialized.

Which of the following best describes a valid relationship?

(A) An instance of the `vehicle` class is `Car`.

(B) An instance of the `Car` object is `vehicle`.

(C) An attribute of the `year` object is `int`.

(D) An attribute of the `vehicle` object is `color`.

117. A student has created an `OrderedPair` class to represent points on an x y-plane. The class contains the following.

- A `double` variable named `x` to represent the x-coordinate of a point
- A `double` variable named `y` to represent the y-coordinate of a point
- A method named `printXY` that prints the values of `x` and `y`

The object `origin` is declared as type `OrderedPair`.

Which of the following statements is true?

(A) `origin` is an instance of the `OrderedPair` class.

(B) `origin` is an instance of two `double` objects.

(C) `OrderedPair` is an instance of the `origin` object.

(D) `printXY` is an instance of the `OrderedPair` class.

118. A student has created a `Song` class. The class contains the following three variables.

- A `String` variable named `artist` to represent the artist name
- A `String` variable named `title` to represent the song title
- A `String` variable named `album` to represent the album title

The object `happyBirthday` will be declared as type `Song`.

Which of the following statements is true?

(A) `artist`, `title`, and `album` are instances of the `Song` class.
(B) `happyBirthday` is an instance of three `String` objects.
(C) `happyBirthday` is an instance of the `Song` class.
(D) `Song` is an instance of the `happyBirthday` object.

119. Consider the following class declaration.

```
public class Sample
{
private int a;
private double b;

public Sample(int x, double y)
{
    a = x;
    b = y;
}

// No other constructors
}
```

The following method appears in a class other than `Sample`.

```
public static void test()
{
Sample object = new /* missing constructor call */ ;
}
```

Which of the following could be used to replace `/* missing constructor call */` so that the method will compile without error?

(A) `Sample()`

(B) `Sample(int x = 10, double y = 6.2)`

(C) `Sample(int x, double y)`

(D) `Sample(10, 6.2)`

120. Consider the following class definition.

```
public class Student
{
  private String myName;
  private int myAge;

  public Student()
  { /* implementation not shown */ }

  public Student(String name)
  { /* implementation not shown */ }

  public Student(String name, int age)
  { /* implementation not shown */ }

  // There are no other constructors.
}
```

Which of the following statements will cause a compilation error?

(A)
```
Student a = new Student();
```

(B)
```
Student b = new Student("Juan");
```

(C)
```
Student c = new Student("Juan", 15);
```

(D)
```
Student d = new Student("Juan", "15");
```

121. The `WindTurbine` class has two constructors. The constructor headers are shown.

```
public WindTurbine(int efficiency)

public WindTurbine(int efficiency, String location)
```

Which of the following statements, when placed in a method in a class other than `WindTurbine`, will construct a `WindTurbine` object `wt`?

(A)
```
WindTurbine wt = new WindTurbine(3);
```

(B)
```
WindTurbine wt = new WindTurbine(3.25);
```

(C)
```
WindTurbine wt = WindTurbine(3, "Atlantic Ocean");
```

(D)
```
WindTurbine wt =  WindTurbine(3.25, "Atlantic Ocean");
```

122. The Player class has only one constructor. A partial declaration of the constructor is shown.

```
public Player(boolean isOnline, int numLives)
{ /* implementation not shown */ }
```

Which of the following statements correctly creates a Player object?

(A)
```
Player p = new Player(true, 5);
```

(B)
```
Player p = new Player(5, true);
```

(C)
```
Player p = Player(true, 5);
```

(D)
```
Player p = Player(5, true);
```

123. The Person class has exactly two constructors. Partial declarations of the constructors are shown.

```
public Person(int idNumber, boolean isActive)
{ /* implementation not shown */ }

public Person(int idNumber, int age, boolean isActive)
{ /* implementation not shown */ }
```

Which of the following statements does **not** correctly create an object of type Person?

(A)
```
Person p1 = new Person(1138, true);
```

(B)

Person p2 = new Person(3136, 17, false);

(C)

Person p3 = new Person(3544, false, 18);

(D)

Person p4 = new Person(0, false);

124. Consider the following class definition.

```
public class Bird
{
private String species;
private String color;
private boolean canFly;

public Bird(String str, String col, boolean cf)
{
    species = str;
    color = col;
    canFly = cf;
}
}
```

Which of the following constructors, if added to the `Bird` class, will cause a compilation error?

(A)

```
public Bird()
{
species = "unknown";
color = "unknown";
canFly = false;
}
```

(B)

```
public Bird(String col, String str)
{
species = str;
color = col;
canFly = false;
}
```

(C)
```
public Bird(boolean cf, String str, String col)
{
 species = str;
 color = col;
 canFly = cf;
}
```

(D)
```
public Bird(String col, String str, boolean cf)
{
 species = str;
 color = col;
 canFly = cf;
}
```

125. Consider the following Point2D class constructor header.

```
public Point2D(double x, double y)
{ /* implementation not shown */ }
```

Which of the following code segments will correctly create an instance of a Point2D object?

(A)
```
Point2D p = Point2D(3.0, 4.0);
```

(B)
```
new p = Point2D(3.0, 4.0);
```

(C)
```
new Point2D = p(3.0, 4.0);
```

(D)
```
Point2D p = new Point2D(3.0, 4.0);
```

126. The `Sofa` class has only two constructors. Partial declarations of the two constructors are shown.

```
public class Sofa(int numOfSeats, double length, String color)
{ /* implementation not shown */ }

public class Sofa(int numOfSeats, double length)
{ /* implementation not shown */ }
```

Which of the following declarations, appearing in a class other than `Sofa`, will correctly create a `Sofa` object?

(A)
Sofa s1 = new Sofa(3, 7.5);

(B)
Sofa s2 = new Sofa(3.0, 8.0);

(C)
Sofa s3 = new Sofa("3", "7.5", "brown");

(D)
Sofa s4 = new Sofa(3.0, 8.0, "blue");

127. The Vbox class includes the following two constructors.

Constructor Signature	Explanation
public vBox(int w, int h, int d)	Constructs a Vbox object that represents a box with width w, height h, and depth d.
public vBox(int len)	Constructs a Vbox object that represents a box with width len, height len, and depth len.

Which of the following declarations, appearing in a class other than Vbox, will correctly instantiate a Vbox object?

(A)
Vbox b1 = new Vbox(4.0);

(B)
Vbox b2 = new Vbox(4, 2);

(C)
Vbox b3 = new Vbox(2, 8, 4);

(D)
Vbox b4 = new Vbox(4.0, 4.0, 4.0);

128. The `Book` class includes the following method.

Method Signature	Explanation
`public int getNumPages(int ch)`	Returns the number of pages in chapter number `ch`

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Assume that `Book` object `book1` and `int` variable `c` have been properly declared and initialized in a class other than `Book`. Which of the following statements will compile without error?

(A)

```
int pages = getNumPages(Book book1, int c);
```

(B)

```
int pages = getNumPages(book1, c);
```

(C)

```
int pages = book1.getNumPages(int c);
```

(D)

```
int pages = book1.getNumPages(c);
```

129. The Dog class contains a non-static method named calculateWalkLength which can be called from a class other than Dog. The method takes an integer parameter for the number of minutes a dog is taken for a walk and returns a double representing the number of miles walked during that time.

The following code segment appears in a class other than Dog. The statement in line 2 causes a compile-time error.

```
Dog fido = new Dog();                    // line 1
double miles = calculateWalkLength(fido, 90); // line 2
```

Which of the following can be used to replace line 2 so that this code segment will compile without error?

(A)

```
double miles = calculateWalkLength(90);
```

(B)

```
double miles = Dog.calculateWalkLength(fido, 90);
```

(C)

```
double miles = Dog.calculateWalkLength(90);
```

(D)

```
double miles = fido.calculateWalkLength(90);
```

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130. The Game class includes the following method.

Method Signature	Explanation
public void addPoints(int pts)	Increases the number of points by `pts`

Assume that the `Game` object `myGame` and the `int` variable `points` have been properly declared and initialized in a class other than `Game`. Which of the following code segments is a valid method call?

(A)

```
myGame.addPoints(int points);
```

(B)

```
myGame.addPoints(points);
```

(C)

```
addPoints(Game myGame, int points);
```

(D)

```
addPoints(myGame, points);
```

131. Consider the following information about the Person class.

- The class has an int attribute age that represents a person's age.
- The class has a non-static getAge method that has no parameters and returns the value of age. This method can be called from another class.

The following code segments each appear in a class other than Person. Assume that p is a Person object whose age attribute is equal to 16.

Code Segment 1

```
int val1 = p.getAge();
System.out.println(val1);
```

Code Segment 2

```
int val2 = Person.getAge();
System.out.println(val2);
```

What, if anything, is printed as a result of executing each of the code segments?

(A) Both code segments print 16.
(B) Code segment 1 prints 16, but code segment 2 prints nothing because it will not compile.

(C) Code segment 2 prints 16, but code segment 1 prints nothing because it will not compile.
(D) Neither code segment will print anything because they will not compile.

132. Consider the following information about the `Triangle` class.

- The `Triangle` class contains three attributes, `side1`, `side2`, and `side3`, to represent the lengths, in centimeters, of each side of a triangle.
- The class contains one constructor that has three `double` parameters that are used to initialize the attributes of a triangle.
- The class contains an instance method, `perimeter`, which takes no parameters. The method prints the perimeter of a triangle followed by a space (`" "`).

The following code segment appears in a class other than `Triangle`.

```
Triangle isosceles = null;
Triangle scalene = new Triangle(3.0, 4.0, 5.0);
Triangle right = scalene;

isosceles.perimeter();
scalene.perimeter();
right.perimeter();
```

What, if anything, is printed as a result of executing the code segment?

(A)
0.0 12.0

(B)
0.0 12.0 12.0

(C) Nothing is printed because `right` has not been created using the `new` keyword.
(D) Nothing is printed because a `NullPointerException` occurs.

133. Consider the following code segment.

```
String oldStr = "ABCDEF";
String newStr = oldStr.substring(1, 3) + oldStr.substring(4);
System.out.println(newStr);
```

What is printed as a result of executing the code segment?

(A) `ABCD`
(B) `BCDE`
(C) `BCEF`
(D) `BCDEF`

134. Consider the following class definition.

```
public class Beverage
{
 private int temperature;
 public Beverage(int t)
 {
    temperature = t;
 }
 public int getTemperature()
 {
    return temperature;
 }
 public boolean equals(Object other)
 {
    if (other == null)
    {
      return false;
    }
    Beverage b = (Beverage) other;
    return (b.getTemperature() == temperature);
 }
}
```

The following code segment appears in a class other than `Beverage`. Assume that `x` and `y` are properly declared and initialized `int` variables.

```
Beverage hotChocolate = new Beverage(x);
Beverage coffee = new Beverage(y);
boolean same = /* missing code */;
```

Which of the following can be used as a replacement for `/* missing code */` so that the `boolean` variable `same` is set to `true` if and only if the `hotChocolate` and `coffee` objects have the same temperature values?

(A) `(hotChocolate == coffee)`

(B) `hotChocolate.equals(coffee)`

(C) `hotChocolate.equals(coffee.getTemperature())`

(D) `hotChocolate.getTemperature().equals(coffee.getTemperature())`

135. Consider the following code segment.

```
String temp = "comp";
System.out.print(temp.substring(0) + " "
        + temp.substring(1) + " "
```

```
        + temp.substring(2) + " "
        + temp.substring(3));
```

What is printed as a result of executing the code segment?

(A)
c o m p

(B)
comp com co c

(C)
comp omp mp p

(D)
comp comp comp comp

136. Consider the following code segment.

```
String str = "CompSci";
System.out.println(str.substring(0, 3));
int num = str.length();
```

What is the value of `num` after the code segment is executed?

(A)
3

(B)
4

(C)
6

(D)
7

137. Consider the following code segment.

```
String str = "0";
str += str + 0 + 8;
System.out.println(str);
```

What is printed as a result of executing the code segment?

(A) `08`

(B) `008`

(C) `0008`

(D) Nothing is printed, because numerical values cannot be added to a `String` object.

138. Consider the following code segment.

```
int one = 1;
int two = 2;
String zee = "Z";
System.out.println(one + two + zee);
```

What is printed as a result of executing the code segment?

(A) `12Z`
(B) `3Z`
(C) `12zee`
(D) `3zee`

139. The following code segment is intended to assign to newWord the result created by removing the first occurrence of "a" from word. Assume that the String variable word has been properly declared and initialized. This code segment works for some, but not all, values of word.

```
int aLoc = word.indexOf("a");
String newWord = word.substring(0, aLoc) + word.substring(aLoc + 1);
```

Which of the following conditions best describes the condition in which this code segment will not work as intended and will result in a runtime error?

(A) "a" is the first character in word.
(B) "a" is the last character in word.
(C) word does not contain the character "a".
(D) word contains only a sequence of multiple "a" characters.

140. Which of the following code segments can be used to set the value of the string `str` to `"Good morning, sunshine!"`?

I.
```
String str = "Good " + "morning," + " sunshine!";
```

II.
```
String str = "Good";
str += " morning, sunshine!";
```

III.
```
String str = " morning, ";
str = "Good" + str + "sunshine!";
```

(A) I only
(B) II only
(C) II and III only
(D) I, II, and III

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141. Consider the following code segment.

```
String one = "xyz";
String two = one.substring(2) + one.substring(1)
        + one.substring(0);
int pos = two.indexOf("x");
```

What is the value of pos after executing the code segment?

(A)
-1

(B)
2

(C)
3

(D)
5

142. Consider the following code segment, which is intended to create a String that consists of the last character in word and assign it to the variable lastChar.

```
String word = /* initialization not shown */;  // line 1
int len = word.length();                        // line 2
String lastChar = word.substring(len);          // line 3
```

Which of the following best describes why this code segment will not work as intended?

(A) substring needs to be called with two parameters, so the method call in line 3 should be changed to word.substring(len, len + 1).

(B) len is not the correct index of the last character in word, so the method call in line 3 should be changed to word.substring(len - 1).

(C) len is not the correct index of the last character in word, so the method call in line 3 should be changed to word.substring(len + 1).

(D) substring cannot be called with a variable as the parameter, so the method call in line 3 should be changed to word.substring(word.length()).

143. Consider the code segment below.

```
int a = 1988;
int b = 1990;

String claim = " that the world's athletes " +
           "competed in Olympic Games in ";
```

```
String s = "It is " + true + claim + a +
        " but " + false + claim + b + ".";

System.out.println(s);
```

What, if anything, is printed when the code segment is executed?

(A) `It is trueclaim1998 but falseclaim1990.`

(B) `It is true that the world's athletes competed in Olympic Games in a but false that the world's athletes competed in Olympic Games in b.`

(C) `It is true that the world's athletes competed in Olympic Games in 1988 but false that the world's athletes competed in Olympic Games in 1990.`

(D) Nothing is printed because the code segment does not compile.

144. Consider the following code segment.

```
String oldStr = "ABCDEF";
String newStr = /* missing code */;
System.out.println(newStr);
```

Which of the following expressions can be used to replace /* missing code */ so that the code segment prints the string "BCEF"?

(A)
oldStr.substring(1, 2) + oldStr.substring(4)

(B)
oldStr.substring(1, 2) + oldStr.substring(5)

(C)
oldStr.substring(1, 3) + oldStr.substring(4)

(D)
oldStr.substring(1, 3) + oldStr.substring(5)

145. In the following code segment, `mon` and `tue` are properly declared `String` variables.

```
String str1 = mon.substring(0, mon.indexOf(" "));
String str2 = tue.substring(tue.indexOf(" ") + 1);
System.out.println(str1 + " and " + str2);
```

For which of the following values of `mon` and `tue` will the code segment print `"cloudy and foggy"`?

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(A)

mon	tue
"cloudy today"	"foggy tomorrow"

(B)

mon	tue
"cloudy today"	"tomorrow foggy"

(C)

mon	tue
"today cloudy"	"foggy tomorrow"

(D)

mon	tue
"today cloudy"	"tomorrow foggy"

146. Consider the following code segment.

```
String original = "DEFGHI";
String result = /* missing code */;
System.out.println(result);
```

Which of the following expressions can be used to replace /* missing code */ so that this code segment prints the string "HIDE"?

(A)

```
original.substring(4) + original.substring(0, 1)
```

(B)

```
original.substring(4) + original.substring(0, 2)
```

(C)

```
original.substring(5) + original.substring(1, 2)
```

(D)

```
original.substring(5) + original.substring(1, 3)
```

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147. Consider the `processWords` method. Assume that each of its two parameters is a `String` of length two or more.

```
public void processWords(String word1, String word2)
{
String str1 = word1.substring(0, 2);
String str2 = word2.substring(word2.length() - 1);
String result = str2 + str1;
System.out.println(result.indexOf(str2));
}
```

Which of the following best describes the value printed when `processWords` is called?

(A) The value `0` is always printed.
(B) The value `1` is always printed.
(C) The value `result.length() - 1` is printed.
(D) A substring containing the last character of `word2` is printed.

148. Consider the following code segment.

```
String str1 = "LMNOP";
String str2 = str1.substring(3);
str2 += str1.substring(2, 3);
```

What is the value of str2 after executing this code segment?

(A)
"OPN"

(B)
"OPNO"

(C)
"NOPM"

(D)
"NOPMN"

149. Consider the following code segment.

```
String one = "computer";
String two = "science";
String concat = /* missing code */;
System.out.println(concat);
```

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Which of the following expressions can be used to replace `/* missing code */` so that the code segment prints the string `"pun"`?

(A)

```
one.substring(3, 4) + two.substring(4, 4)
```

(B)

```
one.substring(3, 5) + two.substring(4, 5)
```

(C)

```
one.substring(4, 5) + two.substring(5, 5)
```

(D)

```
one.substring(4, 6) + two.substring(5, 6)
```